Subham Agrawal

B.Tech, Civil and Infrastructure Engineering
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SUMMARY

Final-year IIT Jodhpur undergraduate with a minor in Mathematical and Computational Economics. Builds option pricing, hedging and stock-selection models in Python and tests every result out of sample. Seeking quant research and trading roles.

EDUCATION

Degree	Institute	Year
B.Tech, Civil and Infrastructure Engineering	Indian Institute of Technology Jodhpur	2023 – 2027
Minor, Mathematical and Computational Economics	Indian Institute of Technology Jodhpur	2026 – 2027

Coursework: Probability, Statistics and Stochastic Processes; Econometrics Lab; Games and Economic Behavior (Grade A); DSA; *ongoing:* Financial Derivatives and Risk Management, ML for Economics, Intro to Machine Learning

PROJECTS

Deep Hedging of Options 2026
Independent project | *Python, PyTorch, Monte Carlo* Live app | GitHub

- Trained a neural network to hedge a short call under trading costs, on Heston paths calibrated to SPX with my own pricing engine.
- On 100,000 unseen paths, cut tail loss (CVaR 95%) by 33% vs a Heston delta and variance-swap hedge; with costs, it learned to trade less than half as much as a delta hedge.
- Held up on 16 years of real NIFTY 50 paths: 13% lower tail loss than Black-Scholes delta at 20 bp costs.

Machine Learning Stock Selection on the Nifty 100 2026
Independent project | *Python, LightGBM, scikit-learn* Live app | GitHub

- Ranked Nifty 100 stocks by next-month return and backtested a long-short portfolio net of costs (2010–26).
- Showed that shuffled cross-validation inflated the Sharpe ratio 7 times (3.68 vs 0.53 walk-forward) and found a Sharpe of 5.5 in pure random walks; purged cross-validation removed the leak.

Volatility Dynamics and the Fear Index: Nifty 50 and India VIX 2026
Team project, Financial Derivatives and Risk Management | *Python, pandas, statsmodels* Live app | GitHub

- Analysed 1,536 days of Nifty 50 and India VIX (2020–26): implied volatility exceeded realised on 78% of days (Newey-West $t = 2.8$), a persistent variance risk premium.
- Backtested Black-Scholes-priced protective puts: a VIX-timed hedge cut maximum drawdown from −38% to −22%, most of it in the March 2020 crash.

Landslide Susceptibility Mapping using GIS and Machine Learning 2026
B.Tech Project under Prof. Susmita Panda | *Python, scikit-learn, XGBoost* Live app | GitHub

- Built a landslide-risk model on 4,431 events and 14 satellite-derived factors. Random splits leaked (AUC 0.97 vs 0.85 on unseen regions), so the final SVM was tuned on spatial cross-validation (AUC 0.905).

COMPETITIONS AND CERTIFICATIONS

Inter IIT Tech Meet 13.0 (Zelta Labs) 2024
Quant Team Member, IIT Jodhpur Predictive Modelling, Time Series

- Built crypto price-direction models for IIT Jodhpur's team; the team ranked 11th overall at the national Inter IIT Tech Meet.

VC Analyst Case Study (Fin Maverick) *team ranked 4th of 159; picked 3 of 20 startups for a USD 10M fund* 2026
Quantitative Finance Cohort'24 *Quant Delegate:* stochastic calculus, options pricing, volatility models (Certificate) 2024

ACHIEVEMENTS

- **IOQM 2020:** top 300 in Odisha (Certificate). **JEE Main 2023:** top 1% of 1.3 million candidates. **CBSE Class X (2020):** 92%.
- **WorldQuant Challenge 2024:** Bronze level (Certificate). **IICPC 2024:** Competitive Programming Summer Camp (Certificate).

TECHNICAL SKILLS

- **Probability and Statistics:** conditional probability, expectation, hypothesis tests, regression, time series
- **Derivatives:** Black-Scholes, Greeks, delta hedging, Heston model, Monte Carlo, implied vs realised volatility
- **Programming:** Python (pandas, NumPy, SciPy, statsmodels, scikit-learn, PyTorch), SQL, C++

POSITIONS OF RESPONSIBILITY

Assistant Head of Finance, Prometeo'25 *budgeting for events and artists, about INR 15 lakh of an INR 40 lakh+ budget* 2025
Head of Media, Prometeo'26 *led 25+ juniors; several reels crossed 500k views* 2025 – 2026
Cultural Captain, Inter IIT Cultural Meet 7.0 and 8.0 *led a 25+ member contingent; top-10 finishes* 2024 – 2025